

Positional Mechanisms in Attention:

two factors, one scan, and where the arena’s mechanisms sit

Abstract

Attention has no intrinsic notion of order, and every positional mechanism is a choice about how to put one in. This review organises the current arena — RoPE, ALiBi, xPos, PoPE, CoPE, MapFormer, Selective RoPE, CARoPE, GRAPE, FoX, Jordan-RoPE, LieRE, PaTH, DeltaNet, RWKV-7, Mamba and its successors, MesaNet, Titans, TAPE and TEM — around a single classification and eight axes of variation.

The classification is sharp. If the positional map is linear in the query and key, translation-invariant, and continuous, the family of interval operators is a one-parameter matrix group $A(d) = \exp(dM)$, and classifying positional encodings reduces to classifying the generator M up to conjugacy. Diagonalising M gives exactly two live slots — a real part that scales and an imaginary part that rotates — because $\mathbb{C}$ has exactly two. Defective generators add polynomial factors and nothing else. Everything that appears to escape this picture escapes by breaking one of the three assumptions, and it is usually possible to say which.

Writing the two slots as one complex log-polar accumulator makes the whole arena a choice of where content enters — instantaneously or accumulated, in the magnitude or in the phase — and with what sign and rank. Two of those axes turn out to be where the differences actually live. We give the algebra, the relational map, and the measurements that separate the axes on a navigation task where they are distinguishable.

1 The shared frame

Rotary attention transforms the query and key of head h by a block-diagonal rotation before the inner product. Splitting a head’s d_{head} coordinates into $n_b = d_{\text{head}}/2$ planes and identifying each plane with $\mathbb{C}$, write $\tilde{q}_t^{(c)}, \tilde{k}_s^{(c)} \in \mathbb{C}$ for the query at position t and key at position s in plane c . The pre-softmax logit is

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=1}^{n_b} \mu_{t,c}^q \mu_{s,c}^k \cos(\theta_s^{(c)} - \theta_t^{(c)} + \psi_{s,c}^k - \psi_{t,c}^q), \quad (1)$$

in polar coordinates $\tilde{q} = \mu^q e^{i(\psi^q + \theta_t)}$, $\tilde{k} = \mu^k e^{i(\psi^k + \theta_s)}$. Every logit is a sum of products $\mu \cos \varphi$.

A positional mechanism is a choice about one of those two factors. RoPE, MapFormer, Selective RoPE, CARoPE and Mamba-3’s rotation set the phase φ . PoPE, ALiBi, the Forgetting Transformer and every gated linear RNN set the magnitude μ . xPos, MapPoPE and Mamba-3 set both. That is the whole design space of (1), and §2 explains why it is not an artefact of the parameterisation.

2 The classification, and why there are two slots

Let $\tilde{q}_t = F(t)q_t$ and $\tilde{k}_s = G(s)k_s$, so the logit is $q_t^\top F(t)^\top G(s)k_s$, and impose three conditions:

- **Linearity.** F and G act linearly on q and k .
- **Translation invariance.** $F(t)^\top G(s)$ depends only on $s - t$.
- **Continuity** in the interval.

Normalise so $F(t)^\top G(t) = I$, absorbing the constant into the keys. Then $A(d) := F(0)^\top G(d)$ satisfies $A(0) = I$ and $A(d)A(e) = A(d + e)$: the interval operators form a *one-parameter matrix group*, so $A(d) = \exp(dM)$ for a fixed generator M , and the classification of positional encodings is the classification of M up to conjugacy. This argument is due to Puranik (2026); GRAPE (Zhang et al., ICLR 2026) develops the same framework independently and uses it to prove exact special cases.

Putting M in Jordan form gives three cases and no others.

- **Diagonalisable, real eigenvalue λ .** The plane contributes $e^{\lambda d}$: exponential decay for $\lambda < 0$, NoPE at $\lambda = 0$, and a blow-up for $\lambda > 0$ that no causal design wants.
- **Diagonalisable, conjugate pair.** The real canonical form is $e^{\lambda d} \text{rot}(\omega d)$ — *damped RoPE*. This is the Re / Im split of (1), and **there are exactly two slots because $\mathbb{C}$ has exactly two**. RetNet, xPos and Mamba-3 all live here.
- **Defective.** Repeated eigenvalues with a nilpotent part give polynomial factors d^r times the above. ALiBi is realisable this way after augmenting q and k ; Jordan-RoPE (2026) is the deliberate occupant, coupling a complex eigenvalue to a nilpotent in the same block to obtain modes $d^r e^{i\omega d}$.

A remark of Puranik’s is worth isolating because it is the bridge to everything content-dependent: a gate with a data-dependent decay should be read as *learning how far to advance time*, not as changing a rate. That is exactly the substitution $t \mapsto S_t = \sum_{u \leq t} G(x_u)$ of §5.

3 The phase axis: a family of scans

Every content-dependent phase mechanism in the arena maintains a state by cumulative summation and reads it out linearly:

$$s_t = s_{t-1} + \varphi(x_{t-K+1}, \dots, x_t) \in \mathbb{R}^m, \quad \theta_t = A s_t \in \mathbb{R}^{Hn_b}, \quad (2)$$

and they differ in three numbers: the **state rank** m , the form of the increment φ , and its **temporal support** K .

mechanism	m	φ	K	readout A
RoPE	1	$\equiv 1$ (constant)	1	ω
MapFormer	r	$W_{\text{out}} W_{\text{in}} x$	1	$\text{diag}(\omega)$
CARoPE	H	$1/(\text{softplus}(xW) + 1) \in (0, 1)$	1	geometric
Selective RoPE	Hn_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	4	τI

RoPE is the $\varphi \equiv 1$ special case: $\theta_t = t\omega$ is a clock that advances by the same amount on every token. Everything else lets the content set the rate.

3.1 Nesting

RoPE $\subset$ MapFormer $\subset$ Selective RoPE, each inclusion *removing a constraint* rather than adding a mechanism.

MapFormer inside Selective RoPE. Four settings suffice: (i) the convolution kernel becomes a delta at the last tap, so K collapses to 1; (ii) the gate bias is set to a constant, making $\sigma(\cdot)$ a scalar g ; (iii) the angle projection is set to $A/(\tau g)$ where $A = \text{diag}(\omega)W_{\text{out}}W_{\text{in}}$; (iv) the projection bias is zeroed. The two models then compute the same θ to 1.5×10^{-5} in float32, and the constructed W_ω has rank 2 of 64.

RoPE inside Selective RoPE is unconditional: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t+1)b$, an exact index clock (constant to 7×10^{-9}).

RoPE inside MapFormer costs something, because φ has no bias term: a constant increment requires every embedding row to share its projection onto W_{in} . Taking $W_{\text{in}} = [I_r \ 0]$ and pinning the first r embedding coordinates achieves it exactly, at a price of r of the d embedding dimensions.

Consequently **every comparison inside this column asks whether a constraint is worth keeping, and none of them is about expressivity.**

3.2 The factorisation identity

Cumulative summation is linear, so it commutes with the readout:

$$\theta_t = \omega \odot \sum_{u \leq t} W_{\text{out}} W_{\text{in}} x_u = \text{diag}(\omega) W_{\text{out}} \sum_{u \leq t} W_{\text{in}} x_u. \quad (3)$$

All $Hn_b = 64$ of MapFormer’s phase channels are linear readouts of a single r -dimensional accumulator (checked against the implementation at 1.5×10^{-5} in float32). The multi-scale frequency ladder is a *view* of one quantity, not 64 independent clocks — which is what makes the phase a coherent position code rather than a bag of unrelated oscillators.

3.3 Coherence rank

Define the **coherence rank** ρ as the dimension of the affine span of $\{\theta_t\}$, i.e. $\text{rank}[A \cdot B \mid A\varphi(0)]$ when $\varphi(x) = Bx$. The affine part matters: RoPE’s constant φ has $B = 0$, so the linear rank alone would read 0. Then $\rho = 1$ for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE.

Two thresholds bracket the useful range. Below, a packing argument: if N^D positions are carried by a ρ -dimensional code, the minimum separation of distinct positions falls at worst like

$$\min_{p \neq p'} \|\theta_p - \theta_{p'}\| \lesssim N^{-\max(D/\rho, 1)}, \quad (4)$$

saturating at the lattice’s own spacing once $\rho \geq D$. This is an upper bound and is not tight: measured exponents track it at $D \leq 3$ and run steeper at $D = 5$ (-2.99 against a predicted -2.50). Above, once $\rho \geq D$ the bound says nothing further, and what remains is a question about conditioning rather than capacity — see §11.

3.4 What $K > 1$ costs

With $K = 1$ and φ linear, (2) is exactly the action of the free *commutative* monoid on tokens: the phase depends on the multiset of tokens traversed and nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a)$ and $\prod_{u \leq t} R(\omega \Delta_u) = R(\omega \sum_{u \leq t} \Delta_u)$. That collapse is what buys the parallel prefix scan.

A convolution with $K > 1$ makes a token’s increment depend on its neighbours, so the phase stops being a function of the multiset and the map from action sequences to $SO(2)^{n_b}$ is no longer a homomorphism. Commutativity is the central computational property here, and it is also the stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable by a fixed per-token increment.

4 The magnitude axis

The other factor of (1). Three occupants, differing in whether content enters instantaneously or by accumulation.

PoPE (Gopalakrishnan et al., 2025) is defined by *deleting* a term. RoPE’s real-valued q carries an implicit intrinsic phase ψ^q , and (1) shows it entering the cosine alongside the positional angle — what PoPE calls an entanglement of “what” with “where”. PoPE removes the interaction $\psi^k - \psi^q$ and keeps only a content-dependent *magnitude*, $\mu = \text{softplus}(\cdot) \geq 0$, with the phase a pure index clock plus a learned per-(head, frequency) constant δ_c . It is the exact complement of MapFormer, which amplifies the same phase term into a learned accumulator.

Forget gates. FoX computes a scalar $f_t = \sigma(w_f^\top x_t + b_f)$ per head and adds $\sum_{j < \ell \leq t} \log f_\ell$ to the logit. Gated linear RNNs (GLA, Mamba, Mamba-2) do the same thing inside a recurrence, $S_t = \text{Diag}(\alpha_t)S_{t-1} + k_t v_t$. GRAPE proves FoX is an exact instance of its additive family. All are accumulated content-dependent magnitude, with $\log \gamma < 0$ by construction.

ALiBi is the index-driven, content-free member: a linear-in-lag penalty, realisable by a defective generator.

5 One equation

The two axes are disjoint coordinates of a single object. Put each token’s contribution in *log-polar* form. Let $c^q, c^k, G : \mathbb{R}^d \rightarrow \mathbb{C}^{n_b}$ and set

$$\boxed{\tilde{q}_t = \exp(c^q(x_t) - \overline{S_t}), \quad \tilde{k}_s = \exp(c^k(x_s) + S_s), \quad S_t = \sum_{u \leq t} G(x_u),} \quad (5)$$

componentwise, with $a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{Re}[\tilde{q}_t^{(c)} \overline{\tilde{k}_s^{(c)}}]$. Writing $c = \gamma + i\pi$ and $G = \delta + i\eta$ with all four real,

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{e^{\gamma_{t,c}^q + \gamma_{s,c}^k + \sum_{t < u \leq s} \delta_{u,c}}}_{\text{magnitude}} \cos \left(\underbrace{\pi_{s,c}^k - \pi_{t,c}^q + \sum_{t < u \leq s} \eta_{u,c}}_{\text{phase}} \right). \quad (6)$$

c is the instantaneous complex log and G the accumulated one; real parts are magnitude, imaginary parts are phase. Every mechanism in §§3–4 is a choice of which of the four real fields $\gamma, \pi, \delta, \eta$ is non-trivial.

The conjugate is load-bearing, and the two accumulators carry opposite signs. The Hermitian form $\tilde{q} \tilde{k}$ *subtracts* phases and *adds* log-magnitudes. So $\text{Im } G$ must be carried with the *same* sign on q and k to yield $\sum_{t < u \leq s} \eta_u$, while $\text{Re } G$ must be carried *oppositely* to yield $\sum_{t < u \leq s} \delta_u$; carried with the same sign it would give $\sum_{u \leq t} + \sum_{u \leq s}$, a joint scale carrying no information about

separation. Perturbing the prefix $u \leq t$ and asking whether the score moves — a relative score must not:

Re G	Im G	relative prefix leak
opposed	aligned	3.4×10^{-16}
opposed	opposed	1.4
aligned	aligned	3.1
aligned	opposed	7.1

Exactly one of the four combinations is interval-relative, and it is the one (5) writes. With $\delta = \log \gamma < 0$ that slot *is* the forget gate; with η content-dependent the phase slot is MapFormer and Selective RoPE. **MapPoPE is not a third mechanism** — PoPE constrains c , the phase family constrains Im G , the coordinates are disjoint, so imposing both is an immediate consequence rather than a design.

Interval-invariance is the right notion of “relative”. With content-dependent increments the logit is not a function of $s - t$ and the score matrix is not Toeplitz. The property that survives, and the one that matters, is that the score depends on the tokens in $(t, s]$ and not on the prefix.

6 The readout, and what the long-context literature varies

§3 treats the readout $A = \text{diag}(\omega)$ as fixed. A large and practically important literature varies exactly that, and it is disjoint from everything above: Position Interpolation, RoPE scaling laws, NTK-aware scaling, Dynamic NTK, NTK-by-parts, YaRN, LongRoPE and LongRoPE2 all rescale the frequency ladder so a model trained at one length runs at another. Every one of them is index-driven, so on Table 1 they occupy a single cell — but they vary something the eight axes do not name.

The critical dimension. The argument that organises that line is worth importing. Channel r has wavelength $T_r = 2\pi b^{2r/d_{\text{head}}}$. High-frequency channels complete many cycles inside the training length L_0 and are trained through their whole phase range; low-frequency channels may not complete even one. The boundary

$$r_{\text{crit}} = \min\{r : T_r \geq L_0\}$$

separates them, and beyond the training length the channels past it enter phase intervals never seen in training. For LLaMA-2 ($d_{\text{head}} = 128$, $L_0 = 4096$) that is 36 of 128 dimensions. This is a mechanism for length-extrapolation failure that is about *which channels were trained*, not about capacity or precision.

Tested, and the account is wrong. Transposed to a content-dependent phase the coordinate is the accumulator rather than the index, which gives a prediction: the benefit of the OOD-helping mechanisms should localise to the channels whose period exceeds the trained range of S_t , and ablating those channels should cost *less* at extrapolated length because they are already being read out of distribution.

The precondition holds — at the training length 8–17 of 64 channels never complete a cycle of their own period, and at $8\times$ that length only 1–4 do — so the mechanism could operate. It does not. Ablating the 32 lowest-frequency channels is free at training length (-0.001) and costs 0.14–0.25 at $T = 1024$, on every arm whose baseline is off the floor. **Those channels are more**

load-bearing at extrapolated length, not less, which is the opposite of the prediction. Damage is localised in frequency — high-frequency ablation costs 0.42–0.57 — but the plainer reading wins: the low-frequency channels carry position at map scale, and coarse position matters more when the walk covers more of the map.

What the same measurement did find: the accumulator’s growth law. All four arms start from the same accumulator range at training length (88–94) and separate entirely by how fast it grows with sequence length, $\text{range}(S) \sim T^\alpha$:

arm	$T=128$	$T=1024$	α	opposition score
signed, $r=4$	91.8	269.7	0.518	0.125
signed, $r=4$ (second batch)	94.1	279.7	0.524	0.092
signed, $r=2$	88.5	320.5	0.619	0.495
monotone, $r=4$	91.5	649.8	0.943	1.849

A signed, well-conditioned increment gives $\alpha \approx 0.52$: the accumulator performs a **diffusive random walk**. A monotone one gives $\alpha \approx 0.94$: **ballistic drift**. The last column is the opposition score $\|\Delta(+x) + \Delta(-x)\|/\|\Delta\|$, measured in a different batch for a different purpose; across the four arms $r(\text{opposition}, \alpha) = +0.9995$.

This is one mechanism for two separate results. How completely opposite actions cancel sets whether the accumulator diffuses or drifts; that sets how fast it leaves the range it was trained on; and that sets the rate of degradation with length. A skewed rank-2 basis (opposition 0.495) and a monotone increment (1.849) are the same failure at different severities. Four points, correlational across arms; $\alpha \approx 0.5$ for a signed walk is expected in itself, and what carries the result is the spread to 0.94 and the cross-batch agreement with a quantity measured for another reason.

Scope of the existing surveys. This is worth stating plainly because it determines what this review is for. The most recent survey of position encoding (Li, August 2026) is a careful treatment of exactly the line above — absolute and relative methods, RoPE, and long-context scaling — and contains **zero occurrences** of CoPE, MapFormer, Selective RoPE, CARoPE, FoX, Mamba or GRAPE. The 2023 length-extrapolation survey mentions CoPE and Mamba twice each and none of the rest. The content-dependent half of the arena, the memory reading of §8, and the navigation regime are not covered by either; the frequency-allocation and context-extension line is not covered here. The two partitions are close to complementary.

7 The relational map

Eight axes separate everything in the arena. They are not independent, but no two mechanisms agree on all eight.

A1 Factor — magnitude (Re), phase (Im), both, or neither.

A2 Locus — instantaneous (c) or accumulated (G).

A3 Driver — the index, the token content, a known position, or a query–key *pair*.

A4 Algebra — semisimple diagonal ($\mathbb{C}^\times$), unipotent, non-semisimple, or non-abelian.

A5 Sign — may the increment be negative?

A6 Rank of the content-to-increment map.

A7 Host — softmax attention, or linear attention / SSM.

A8 Composition — parallel prefix sum, path-ordered product, or neither.

Every *abelian* row above composes by parallel prefix sum (A8); LieRE and Algebraic PE are the exceptions, since non-commuting generators obey no prefix-sum law.

7.1 What leaves the frame, and how

mechanism	assumption broken	consequence
PaTH, DeltaNet, RWKV-7 CoPE	the <i>fixed basis</i> (A4 non-abelian) <i>per-token factorisation</i> (A3 = pair)	path-ordered product of $I + \text{rank-1}$; no S_t $p_{ij} = \sum_{k=j}^i \sigma(q_i \cdot k_k)$; no θ_t
MesaNet, Titans, TAPE	the <i>path structure</i> (A8)	prefix-global inverse / deep memory / lay- erwise update
Shaw, T5, Transformer-XL	<i>continuity</i> (Puranik’s third)	a lookup table over clipped or bucketed lag; no generator M
TEM-t	<i>linearity</i> and <i>commutativity</i>	$e_t = \sigma(e_{t-1} W_{a_t})$; see §9

7.2 The classical relative methods, and a fourth way out of the frame

Shaw-style relative position representations, Transformer-XL and T5’s relative bias predate the rotary line and are still widely deployed. They belong in the map, and they occupy a slot the out-of-frame table did not have.

Shaw et al. add a learned vector indexed by the *clipped* lag,

$$s_{ij} = \frac{q_i^\top (k_j + a_{r_{ij}}^K)}{\sqrt{d_{\text{head}}}}, \quad r_{ij} = \text{clip}(j - i, -K, K),$$

and T5 compresses the same idea to a per-head scalar, $s_{ij} += b_{h, \text{bucket}(j-i)}$. Transformer-XL carries the relative form across cached segments.

These are linear in q and k and translation-invariant, but the map from lag to operator is a **lookup table over a discretised lag**. That fails Puranik’s third condition, *continuity*, and with it the group law: there is no generator M because clip and bucket are step functions, and beyond the table’s range the operator is constant rather than continuing. So they leave the frame by a route distinct from non-abelian transport, pair-dependence or a broken path structure.

The trade-off is exact and worth stating, because it is the one design choice the rotary line silently makes. **A table can represent an arbitrary function of the lag inside its window and cannot extrapolate outside it at all; a one-parameter group is restricted to the three families of §2 and extrapolates by construction.** RoPE gives up expressiveness over short lags to buy a functional form that is defined at every lag. Whether that purchase is honoured is a separate question, and the long-context literature’s own conclusion is that it frequently is not: the ability to *compute* positional features beyond the training length does not imply reliable generalisation there, which is why context extension is evaluated by short-context retention, position-wise perplexity, retrieval and long-context reasoning rather than by whether the encoding is defined.

7.3 Reading the map

(i) *The frame’s own cells are full.* Crossing A1 (magnitude / phase) with A3 (index / content) gives four cells and all four are occupied: RoPE, ALiBi, Mamba-3’s rotation, FoX. GRAPE holds

Table 1: Mechanisms that fit (5) or its unipotent extension. *Jordan-RoPE’s magnitude factor is a *polynomial* in the lag, $d'' e^{i\omega d}$; its generator couples a complex eigenvalue to a nilpotent in one defective block. †FoX’s bias $\sum \log f_\ell$ and GLA’s weight $\prod \gamma$ are the same object; the unipotent label follows GRAPE’s realisation, which augments q, k to embed the bias in GL , and read as $\mathbb{C}^\times$ it sits with Mamba/GLA. §Scoped: LieRE reads its rotation directly off a known position and its generators need not commute; Algebraic PE is abelian for sequences and grids but not for k -ary trees; GRAPE’s additive family is “rank-1 (or low-rank)” in its own abstract, and “full” applies only to its general edge potential. ¶MapEM evaluates the same equation on the *tensor product* of the content and position spaces rather than on $\mathbb{C}^{n_b}$; see §9.

mechanism	A1 factor	A2 locus	A3 driver	A4 algebra	A5 sign	A6 rank	A7 host
NoPE	—	—	—	—	—	—	softmax
RoPE	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
ALiBi	magnitude	acc.	index	unipotent	fixed < 0	n/a	softmax
xPos	both	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
PoPE	magnitude	inst.	content	—	> 0	full	softmax
MapFormer	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapEM [¶]	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapPoPE	both	inst.+acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
Selective RoPE	phase	acc.	content	$\mathbb{C}^\times$	signed	full (or 8–16)	both
CARoPE	phase	acc.	content	$\mathbb{C}^\times$	$\in (0, 1)$	1/head	softmax
GRAPE-M	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
GRAPE-A / AP	magnitude	acc.	content	unipotent	≥ 0	1–full [§]	softmax
FoX	magnitude	acc.	content	unipotent [†]	< 0	scalar/head	softmax
Jordan-RoPE	both*	acc.	index	non-ss.	n/a	n/a	softmax
LieRE	phase	direct [§]	position	$SO(d)$, non-ab.	n/a	tile $b \times b$	softmax
Algebraic PE	phase	acc.	structure	$\mathbb{C}^\times$ [§]	n/a	n/a	softmax
Mamba / Mamba-2, GLA	magnitude	acc.	content	$\mathbb{C}^\times$	< 0	full/scalar	SSM
Mamba-3	both	acc.	content	$\mathbb{C}^\times$	signed	full	SSM

the index-multiplicative and content-additive cells; Mamba-3, Selective RoPE and CARoPE hold content-phase. The remaining structure is **content-dependent, signed, orthogonal and still abelian** — contextual multiplicative — which is exactly where MapFormer, Selective RoPE and CARoPE sit.

(ii) *A5 is settled next door and unapplied here.* Sarrof et al. (2024) observed that SSM gates are “always nonnegative ... due to exponential or sigmoid parameterizations”, with a theorem: such models cannot recognise PARITY. Grazi et al. (ICLR 2025) proved the eigenvalue version and fixed it by extending the range to $[-1, 1]$; PaTH adopts it deliberately; RWKV-7 restricts it only for training stability; and Selective RoPE’s §4.2 already demonstrates single-layer Transformer parity from rotations that “model *flips*”. Yet CARoPE, GRAPE-AP and CoPE are all monotone — by softplus, by construction, by sigmoid — and none cites any of that literature. §11.1 measures what the constraint costs where it can be seen.

(iii) *A6 is thinner than a blank column suggests.* Two rows constrain something rank-like for reasons of their own: PaTH’s Householder factors are $I + \text{rank-1}$ by definition, and CARoPE emits one scalar per head, putting its content-to-angle map *below* MapFormer’s r rather than above it. The nearest deliberate study is LieRE’s generator tile-size sweep, which varies density on the *output* side and peaks in the interior at 8×8 . What no one sweeps is the *input-side* bottleneck: how many dimensions of content may steer the angle.

(iv) *A7 is nearly free.* Selective RoPE runs in both hosts and Mamba-3 crosses over by state-space duality, so “softmax versus SSM” separates implementations rather than mechanisms.

8 The memory reading: what the frame looks like from the other side

A large literature — fast weight programmers, linear attention, gated linear RNNs, state-space models and their hybrids — studies the same equation while asking a different question. It is worth stating the correspondence, because the two literatures partition the design space almost perfectly and neither says so.

One primitive, two forms. The outer product $v_u \otimes k_u$ binds a value to a key, and superposing bindings gives an associative memory read by a query. The object appears as Kohonen’s correlation-matrix memory (1972), Schmidhuber’s fast weight programmer (1992), the recurrent form of linear attention (Katharopoulos et al., 2020), the hidden state of a scalar- A state-space model, Smolensky’s tensor-product role-filler binding (1990), and — via the dual form of Irie et al. (2022) — the weight matrix of any linearly-parameterised layer trained by gradient descent. Schlag et al. (2021) made the linear-attention correspondence explicit. This gives every mechanism two readings:

$$\underbrace{W_t = \sum_{u \leq t} A(t-u) v_u k_u^\top, \quad y_t = W_t q_t}_{\text{primal: compress, bounded state}} \iff \underbrace{a_{ts} = \tilde{q}_t \cdot \tilde{k}_s \text{ over kept items}}_{\text{dual: keep and attend}}$$

The positional operator is the memory’s transport kernel. $A(d)$ is what happens to a stored trace as it ages d steps, and §2 says there are two things it can do:

slot	positional reading	memory reading
Re G	magnitude / decay	how fast the trace is forgotten
Im G	phase / rotation	how the trace is re-addressed as it ages

It also identifies A8 with the primal/dual choice: **a positional mechanism admits a parallel prefix scan exactly when the memory admits a bounded recurrent state**, because both are the group law $A(d)A(e) = A(d + e)$. That is one condition, not two.

The two literatures have divided the slots without noticing. The fast-weight literature owns $\text{Re } G$ exhaustively. Irie and Gershman’s review (TMLR 2025) categorises the whole modern family by *decay type* — RetNet a constant scalar, Mamba-2 a scalar, GLA a rate per row of the fast weight — and the words “rotary”, “RoPE” and “positional” do not occur in it. The positional-encoding literature owns $\text{Im } G$ and treats decay as a side-condition. Mamba-3 is the first design to claim both deliberately, and it arrives from the memory side, stating the crossing as a proposition on “data-dependent RoPE equivalence”. The hybrid quadratic-linear transformers of Irie et al. (2025) use RoPE, but only as an off-the-shelf component inside the attention half rather than as a design axis.

Positional mechanisms are key-side by construction, which makes the frame a theory of addressing. Gershman, Fiete and Irie (2025) argue that what distinguishes key-value memory from classical similarity-based retrieval is that it holds *separate* representations for storage and for retrieval, so the two can be optimised independently — fidelity for values, discriminability for keys — and that the binding constraint on memory performance is addressing rather than capacity.

Every mechanism in Table 1 acts on q and k and none of them touches v . Read through the correspondence above that is not a convention: **a positional mechanism is an addressing rule**, and $A(d)$ says how the address of a trace changes as it ages. The two slots are then two ways for an address to move: $\text{Im } G$ transports it around a torus, so a trace becomes retrievable again at a predictable displacement; $\text{Re } G$ makes it fade, so the trace becomes unretrievable at a rate the content chooses. Nothing in the frame can alter what was stored, only where it can be found.

Why decay is compulsory on one side of the line and optional on the other. A visible asymmetry in Table 1 is that every SSM row carries $\text{Re } G < 0$ while most softmax rows leave the magnitude slot empty. Schlag et al. (2021) supply the reason: an additive outer-product memory has a finite *capacity*, and once the sequence length exceeds it the model enters an overcapacity regime in which stored associations interfere. In the primal form decay is therefore capacity management and not a modelling preference — something has to evict. In the dual form the items are kept separately and the softmax renormalises, so nothing saturates and a decay term (ALiBi, FoX) is an inductive bias one may decline. The same equation, and the slot is mandatory on one side and free on the other.

That also sharpens what the delta rule is for. Schlag et al. introduce it precisely because purely additive updates cannot *edit* a stale association, only bury it; the rewrite is a capacity fix before it is an expressivity fix, and the expressivity — state tracking, parity — comes along with it.

The non-abelian family is the rewritable-memory family. The first row of the out-of-frame table is not a collection of exceptions. DeltaNet, PaTH and RWKV-7 replace the *additive* Hebbian update with a *delta rule*,

$$W_t = W_{t-1} + \sigma(\beta_t)(v_t - W_{t-1}\phi(k_t)) \otimes \phi(k_t),$$

whose state-transition matrix $I - \sigma(\beta_t)\phi\phi^\top$ is a Householder-like reflection rather than a scalar. Memory becomes rewritable — stale associations are corrected instead of accumulated — and the price is exactly the group law: reflections do not commute, so no $A(d)$ and no S_t exist and the

composition is a path-ordered product. **What that family trades away is the positional structure, and what it buys is a memory that can overwrite.** It is also where the sign axis of §7(ii) is deliberately exercised: with $\sigma = 2$ sigmoid the eigenvalues lie in $(-1, 1)$ *including negative values*, which is what unlocks parity and modular state tracking.

And MapEM’s lift is a binding. The tensor product $q^x \otimes q^p$ of §9 is Smolensky’s role-filler binding: content as filler, position as role. Read from the memory side, MapEM stores each observation bound to its place rather than superposing the two channels additively — which is the same trade the primal/dual axis makes, one level down.

Everything in this section is a change of reading, not of mechanism. It matters because it says where to look for the unoccupied designs: the cells that are empty in one literature are usually the ones the other has already filled.

9 Where MapEM and TEM-t sit

Two mechanisms from the cognitive-map literature are placeable, and the placements are opposite.

MapEM lifts the frame rather than leaving it. MapEM computes two attention matrices and multiplies them elementwise, $A = \text{softmax}(A_X \odot A_P)$. With $(A_X)_{ts} = q_t^x \cdot k_s^x$ and $(A_P)_{ts} = q_t^p \cdot k_s^p$,

$$(A_X \odot A_P)_{ts} = (q_t^x \otimes q_t^p) \cdot (k_s^x \otimes k_s^p),$$

so the Hadamard product *is* a single bilinear form — on the tensor product of the content and position spaces, of dimension $d_x d_p$ rather than d (verified to 1.9×10^{-6} in float32). Everything in §5 applies with c and G valued in $\mathbb{C}^{d_x d_p}$.

That is sharper than “an AND-gate rather than an OR-gate”, and it says what the lift costs: a rank-one tensor $q^x \otimes q^p$ can express content-*and*-position but not content-*or*-position, and there is no fallback channel when either factor is uninformative. It predicts the observed sensitivity to how the position channel is initialised — a single shared p_0 beats separate q_0^p, k_0^p by +0.358 on a matching task (3/3 seeds) — because with no additive fallback a mis-set position factor multiplies the whole score instead of being outvoted.

TEM-t is the same machine with two conditions removed. TEM-t (Whittington et al., 2022) generates position recurrently, $e_t = \sigma(e_{t-1} W_{a_t})$ with σ a ReLU and W_a a general per-action matrix, then sets $Q = K = W_e e$ and $V = W_x x$; content is absent from the logit entirely and enters only through the values.

Its logit is $e_t^\top W_e^\top W_e e_s$. Writing $e_s = W_{t \rightarrow s} e_t$ and $e_t = W_{0 \rightarrow t} e_0$, with W_e orthogonal this is $e_0^\top W_{0 \rightarrow t}^\top W_{t \rightarrow s} W_{0 \rightarrow t} e_0$ — **the interval operator conjugated by the prefix**. Conjugation is trivial exactly when the group is abelian:

transition group	logit depends on the interval alone?
abelian (block rotations), no σ	yes
general $SO(d)$, no σ	no
abelian, with ReLU	no
general $SO(d)$, with ReLU	no

MapFormer is precisely the surviving row: $SO(2)^{nb}$ is abelian by construction and there is no nonlinearity on the path. TEM-t breaks both, and either break alone suffices.

This is the mechanical account of a large empirical cost. No group law means no S_t , hence no parallel prefix scan, hence a sequential loop: at $L = 2048$ a faithful TEM is $120\times$ slower than at $L = 128$ and $1632\times$ slower in absolute terms than a MapFormer of $20\times$ *more* parameters. The constant includes interpreter overhead; the shape is the missing group law. TEM-t is 2022 and MapFormer 2025, so on this axis the later contribution is the *restriction* — abelian and linear — that buys the scan and the relative law.

10 What each mechanism can and cannot do

The axes of §7 are descriptive. This section is the part that predicts: what a task demands of a positional code, which mechanisms supply it, and which trade-offs are forced rather than incidental.

10.1 Three hard trade-offs

(T1) Commutativity buys the scan and costs path-dependence. With $K = 1$ and a linear increment the phase depends on the multiset of tokens traversed, the prefix product collapses to a prefix sum, and inference is $O(\log T)$ parallel. The same property makes displacement-under-accumulated-heading unrepresentable: a fixed per-token increment cannot express “forward” when forward depends on where you are facing. Measured both ways — the position effect falls from $+0.438$ to $+0.050$ under turn/turn/forward actions, and a mechanism that abandons the group law (TEM-t, whose ReLU and non-commuting per-action matrices each break it independently) runs $120\times$ slower at $L = 2048$ than at $L = 128$ with $20\times$ fewer parameters. You may have the scan or the path-dependence.

(T2) A signed increment buys cancellation and is what makes the code survive length. The accumulator’s growth exponent is set by how completely opposite actions cancel: $\alpha \approx 0.52$ (diffusive) when they do, $\alpha \approx 0.94$ (ballistic) when they cannot. A monotone clock leaves its trained range $\sim T^{0.4}$ faster, and at matched training loss it does not beat a plain index clock anywhere. There is no compensating benefit measured on either side; the constraint is pure loss on navigation and invisible on order-only tasks.

(T3) Compressing the memory makes decay compulsory; keeping it makes decay optional and the cost quadratic. In the primal form the outer-product state saturates, so $\text{Re } G < 0$ is capacity management — something must evict. In the dual form items are kept separately and the softmax renormalises, so a decay term is an inductive bias one may decline. And a memory that must *overwrite* stale associations needs a delta-rule update, whose transition matrix is a reflection rather than a scalar — which forfeits T1.

10.2 What a task demands, and what supplies it

if the task needs ...	it requires	and these supply it
<i>order only</i> — which token came first	any monotone clock	RoPE, ALiBi, CoPE, CARoPE, GRAPE-AP, every gated linear RNN
<i>signed displacement</i> — parity, net motion, state tracking	a signed increment	MapFormer, Selective RoPE, Mamba-3; DeltaNet/PaTH/RWKV-7 with $\beta > 1$. Not CARoPE, GRAPE-AP or CoPE
<i>absolute location in a bounded map</i>	signed and well-conditioned ($r \geq 4$)	MapFormer at $r \geq 4$; $r = 2$ degrades because its basis is skewed, not because it lacks capacity
<i>holding position far past the training length</i>	a diffusive accumulator, $\alpha \approx 0.5$	follows from cancellation (T2); or bound the accumulator explicitly, which is what a wrapped filter does
<i>path-dependent actions</i> (turn/forward)	non-abelian transport, or an allocentric recoding of the input	PaTH, DeltaNet, RWKV-7, TEM-t — none evaluated on navigation. Recoding the action to absolute displacement restores +0.488 and keeps the scan
<i>overwriting stale associations</i>	a delta rule	DeltaNet, PaTH, RWKV-7; forfeits the parallel scan
<i>exact recall over a window</i>	the dual form; keep the items	softmax attention. A lossy summary is not a sufficient statistic, which is why hierarchy loses here and wins on compositional transfer
<i>bounded state, linear time</i>	abelian transport	every row of Table 1; measured $2.6\text{--}3.3\times$ over a $16\times$ length increase

10.3 What follows for choosing a model

On language, most of this is invisible. Order is nearly all that is demanded, so a monotone clock suffices, which is why the constraint has gone unnoticed on that side: CARoPE, GRAPE-AP and CoPE are all monotone and none cites the state-tracking literature that proves the limitation. The one place it surfaces on text is parity and formal-language state tracking, where it has been both proved and fixed.

On navigation and any task with a metric structure, the sign and the conditioning of the increment are the whole game. They are worth more per parameter than anything else measured here — the sign costs nothing at all, and $r = 2 \rightarrow 4$ costs 384 parameters — and they dominate the choice of positional encoding by two orders of magnitude (+0.461 for path integration against +0.003 for the encoding).

Where the environment breaks the assumption, fix the environment. No abelian mechanism can represent turn/forward dynamics, and the non-abelian ones pay the scan for it. Recording the absolute displacement instead of the commanded turn is a change to the data, not the model, and it restores the full effect — available wherever heading is known, which is every simulator.

The two literatures are complementary, and the empty cells are informative. The fast-weight line owns the magnitude slot and never touches the phase; the positional line owns the phase and treats decay as a side-condition. Mamba-3 is the first design to claim both. A mechanism that needs a bounded state *and* signed transport *and* rewritability does not currently exist, and T1 says the last two cannot both be had with a parallel scan.

11 What is measured

The axes above are separable only on a task where position must be reconstructed from the input rather than read off the index. All measurements below use a torus grid-navigation task: an agent takes actions on an $N \times N$ torus, observations are aliased across cells, and the model predicts the observation at revisited cells. Standard errors are over seeds; MDE is $2.8 \text{ sd}/\sqrt{n}$, and a contrast inside its MDE is reported as unmeasured.

11.1 The sign of the increment

Six arms $\times$ 12 seeds, one batch, $r = 4$ throughout, **parameter count identical at 204,757** — the constraint adds nothing and does not touch the rank.

arm	increment	$T=128$	$T=512$	$T=1024$	final loss
signed	$W_{\text{out}}W_{\text{in}}x$	1.000	0.978	0.922	0.0002
$ \cdot $	$ W_{\text{out}}W_{\text{in}}x $	0.946	0.675	0.558	0.171
softplus	GRAPE-AP form	0.977	0.798	0.584	0.068
CARoPE	$1/(\text{softplus}(\cdot) + 1)$	0.900	0.809	0.645	0.293
RoPE	index	0.799	0.449	0.345	0.784

At matched training loss, signed path integration beats an index code by +0.123 and +0.195 at $T = 512$ and $T = 1024$ (12/12 seeds); monotone path integration beats an index code nowhere — -0.028 at $T = 128$ and unmeasured beyond. The entire measured value of a content-dependent phase over a plain index clock, on this task, requires the increment to be signed.

The learned code shows why. The opposition score $\|\Delta(+x) + \Delta(-x)\|/\|\Delta\|$ is 0 when opposite actions cancel exactly and 2 when they are identical:

arm	oppose _{x}	oppose _{y}	$ \cos(N, E) $
signed	0.125	0.106	0.218
$ \cdot $	1.849	1.855	0.587
softplus	1.905	1.885	0.723
CARoPE	1.981	1.977	0.930

CARoPE’s published parameterisation reaches 1.98 of a possible 2.00, with its north and east axes all but collapsed onto one another. These models do not merely score worse — they cannot represent a -1 action, and the probe confirms they do not.

Limitation. At the training length the signed baseline is at 1.000 ± 0.000 , so accuracy there cannot show a deficit of any size. The training-length effect is visible in the *loss* instead: every constrained arm fits strictly worse on 12/12 seeds at identical parameters.

Scope. This is a replication in a new regime. The constraint and its consequence for parity are established (§7(ii)); what is new is navigation rather than formal languages, and an isolation in which $|\Delta|$ against Δ is a single operation at identical parameter count.

11.2 Rank

The bottleneck r is meant to be the dimensionality of the action space — for a 2D grid, a 2-vector — which makes $r = 2$ sufficient *by construction*. It is not sufficient in practice. Against $r = 2$ at $8 \times$ the training length ($T = 1024$ against a training length of 128; 8 seeds):

	params added	$T=512$	$T=1024$
$r = 4$	+384	+0.038	+0.085 (8/8, t 3.57)
$r = 8$	+1,152	+0.037	+0.091
$r = 16$	+2,688	+0.028	+0.079
$r = 32$	+5,760	+0.038	+0.095

This is a **step at $r = 2$, not a capacity slope**: flat from $r = 4$ to $r = 32$. The cause is conditioning, not capacity. Given four dimensions the model still places its actions in a 2-plane (100.0% of spectral energy; 99.96% even at $r = 32$), vindicating the dimensional argument — what fails is the *basis* that $r = 2$ is forced into:

	opposition ($N+S \approx 0$)	$ \cos(N, E) $	obs/action norm
$r = 2$	0.495	0.783	0.139
$r = 4$	0.092	0.175	0.042

At $r = 2$ opposite actions fail to cancel by half the action scale and north and east are nearly parallel. The seed standard deviation also falls from 0.064 to 0.012 at $T = 1024$, which matters because it sets every detectable effect size.

Two further observations. The deficit does *not* grow with the dimensionality of the world: at $D = 5$, $r = 2$ is unimpaired (0.896, its best of three conditions), so the packing account of (4) does not explain it. And an unconstrained generator cannot be projected below rank ≈ 16 post hoc while $r = 4$ trains fine — trained-at-rank and truncated-to-rank are different objects.

11.3 What the other constraints are worth

Selective RoPE’s three additions to MapFormer’s generator, measured on the torus at $T = 1024$ and on a parity task at $L = 128$; increments against MapFormer at $r = 2$.

constraint removed	params	parity $L=128$	torus $T=1024$
$K = 1 \rightarrow 4$ (causal conv)	+193	−0.020	−0.064 (t −1.99)
$m = r \rightarrow Hn_b$ (full rank)	+7,873	−0.020	+0.058 (7/8)
φ linear $\rightarrow$ gated	+8,193	−0.030	+0.086 (8/8)
all three	+16,385	−0.009	+0.048 (t 1.36)
$r = 2 \rightarrow 4$, <i>constraint kept</i>	+384	—	+0.085 (8/8)

Two readings. First, **the two $\approx 8k$ -parameter knobs are statistically indistinguishable from each other** (+0.018 and +0.028, both inside their MDEs), so what they buy is most plausibly capacity rather than either named mechanism — and widening the bottleneck MapFormer already has buys the same thing for 384 parameters, $21\times$ cheaper. Second, the convolution is the only knob negative on both tasks, consistent with §3.4: $K > 1$ costs the homomorphism.

Caveat. These arms replace the readout $\text{diag}(\omega)W_{\text{out}}$ by τW_2 . For the conv and gate rows the function class is preserved exactly — it is the construction of §3.1 — so the difference is initialisation (a geometric frequency ladder against a default draw) plus one scalar; those effects are attributable up to initialisation. The two full-rank rows also add weight normalisation and a bias, so their function class genuinely differs.

A separate probe rules out the natural mechanism for the gate. If the sigmoid worked by suppressing observation tokens — which MapFormer must instead learn to send to $\Delta \approx 0$ — the gate would be near zero on observations. It is 0.416, and the action/observation ratio is $1.35\times$ on the torus where the gate helps and $1.54\times$ on parity where it hurts.

11.4 The navigation regime

Every paper in §7 evaluates on language, recall, state tracking, or — for LieRE and Algebraic PE — images, video and trees. None runs grid navigation, and it is the regime in which these axes are distinguishable at all.

The basic separation is large. On a task where the model explores with observations revealed and then continues blind, predicting the observation at each cell, path integration scores 0.730 ± 0.247 ($n = 5, 64^2$) against 0.154 ± 0.018 for an index code, chance 0.0625, with no seed overlap. The axis is path integration and not the encoding: PoPE with path integration scores 0.847 against 0.117 for PoPE with an index. On the standard revisit task the position effect is $+0.461$ against $+0.003$ for the encoding — a factor of 150.

Two boundaries are worth stating. The effect is not a property of navigation as such but of *map extent*: at matched observation aliasing it is -0.010 , $+0.015$ and $+0.305$ for 32, 128 and 512 occupied cells — a threshold, not a gradient. And it does not survive rotation-based action spaces: under turn/turn/forward, where displacement depends on accumulated heading, the effect falls from $+0.438$ to $+0.050$, which is §3.4’s commutativity limitation made concrete. Recording the absolute displacement instead of the commanded turn restores it to $+0.488$.

11.5 Anchors

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
Signed incre- ment	sign of Δ	0	largest per parameter: removing it costs $-0.215/-0.280$ at $T=512/1024$ (12/12)
Bottleneck rank	state rank ρ	+384 (+0.19%)	+0.085 at $T=1024$ for $r=2 \rightarrow 4$ (8/8); flat to $r=32$
PoPE	magnitude μ	+320	+0.027 <i>with</i> path integration; both arms on the floor without it
Recursion	block reuse	0	+0.056 parity (16/16); matches $3\times$ the parameters
Hierarchy	token count	0 if shared	length-dependent: ≈ 0 at $L=16$, $+0.060$ at $L=512$; -22.8% time at $L=2048$

12 Open questions

- **The input-side rank.** $r = 2$ is the worst setting and $r = 4$ fixes it for 384 parameters, but nothing explains why the optimiser finds a skewed basis at $r = 2$ and a well-conditioned one at $r = 4$ when the learned solution occupies a 2-plane either way.
- **The OOD-length axis, now partly explained.** The accumulator’s growth exponent accounts for the sign and rank results together (§6), and it predicts that bounding the accumulator should help at length — which is what a wrapped filter does and what it is measured to do. What it does not yet explain is why the forget gate and PoPE show the same signature, since neither bounds the accumulator. The direct test is to vary α deliberately and check that OOD degradation follows it.

- **Defective generators.** Jordan-RoPE occupies the cell; what polynomial factors in the lag are *for* remains open, and no navigation evaluation exists.
- **Sign on language.** A monotone clock cannot represent a -1 action, and on navigation that costs everything. On text HGRN’s own ablation finds a data-dependent phase worse than a constant one. Whether the sign matters there, and for what, is untested.
- **Non-commuting action spaces.** Rotation-based actions defeat a fixed per-token increment. Allocentric recoding sidesteps it wherever heading is known; the non-abelian mechanisms (PaTH, DeltaNet, RWKV-7) address it directly and have never been evaluated on navigation.

Sources

Every mechanism above was read in its own paper rather than in a summary, and every externally-checkable cell of Table 1 was checked against the source.

<i>the frame</i>		<i>content-dependent phase</i>	
Puranik	Jane Street Blog, Apr 2026	MapFormer	2511.19279
GRAPE	2512.07805, ICLR 2026	Selective RoPE	2511.17388, ICLR 2026
Algebraic PE	2312.16045, NeurIPS 2024	CARoPE	2507.23083
Jordan-RoPE	2605.04217	Mamba-3	2603.15569
		LieRE	2406.10322, ICML 2025
<i>content-dependent magnitude</i>		<i>index-driven</i>	
PoPE	2509.10534	RoPE	2104.09864
FoX	2503.02130	ALiBi	2108.12409
GLA	2312.06635	xPos	2212.10554
Mamba, Mamba-2	2312.00752, 2405.21060	NoPE	2305.19466
HGRN, HGRN2	2311.04823, 2404.07904		
<i>outside the frame</i>		<i>the sign axis</i>	
PaTH	2505.16381	Sarrof et al.	2405.17394
DeltaNet	2406.06484	Grazzi et al.	2411.12537, ICLR 2025
RWKV-7	2503.14456		
CoPE	2405.18719	<i>cognitive maps</i>	
TAPE	2501.00712	TEM-t	2112.04035, ICLR 2022
MesaNet, Titans	2506.05233, 2501.00663		
<i>surveys of the same arena</i>			
Li (position encoding)	2608.10021	Length extrapolation	2312.17044
<i>the memory reading</i>			
Schlag et al.	2102.11174, ICML 2021	Irie et al. (dual form)	2202.05798, ICML 2022
Irie & Gershman	2508.08435, TMLR 2025	Irie et al. (hybrid)	2506.00744, NeurIPS 2025
Gershman et al.	2501.02950, Neuron 2025	Katharopoulos et al.	2006.16236, ICML 2020